



Inspired by Photosynthesis,
Engineered for
Sustainability

SolaForm Artificial Leaf



The need for Sustainable Carbon Capture

BACKGROUND

Industrial Emissions



Factories release large amounts of carbon dioxide into the atmosphere

Transportation



Vehicles powered by fuels emit carbon dioxide every day

Fossil Fuel Combustion



Burning coal, oil and gas increases atmospheric carbon dioxide

PROBLEM

CLIMATE CHANGE



Traps heat in the atmosphere, causing global warming.

Environmental Impact



Extreme weather, rising sea levels, biodiversity loss and pollution

Economic Impact



Damage to infrastructure, agriculture, health and increased cost

SolaForm Artificial Leaf: Converting carbon dioxide into value

A solar powered system that captures carbon dioxide and converts it into formate while producing oxygen

Three operational stages of the SolaForm Artificial Leaf

1. Carbon dioxide Capture

Air is drawn in and carbon dioxide is captured by an amine-based sorbent material

2. Carbon dioxide Release (Regeneration)

The sorbent is heated to release the captured carbon dioxide into the buffer chamber

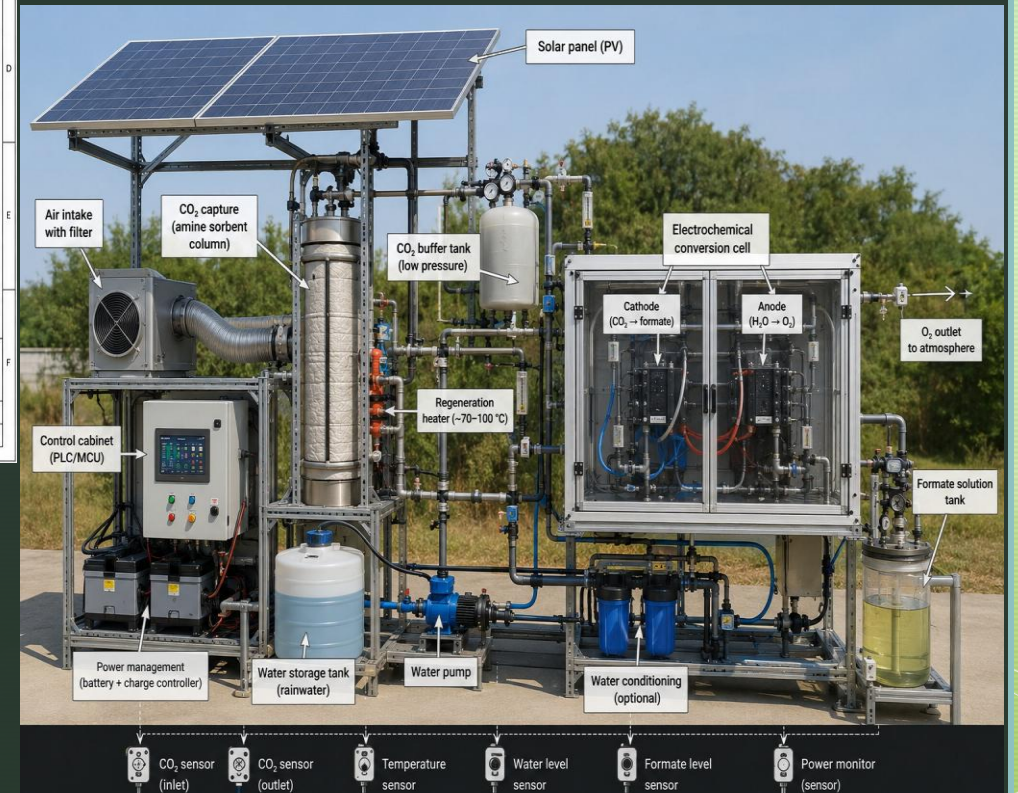
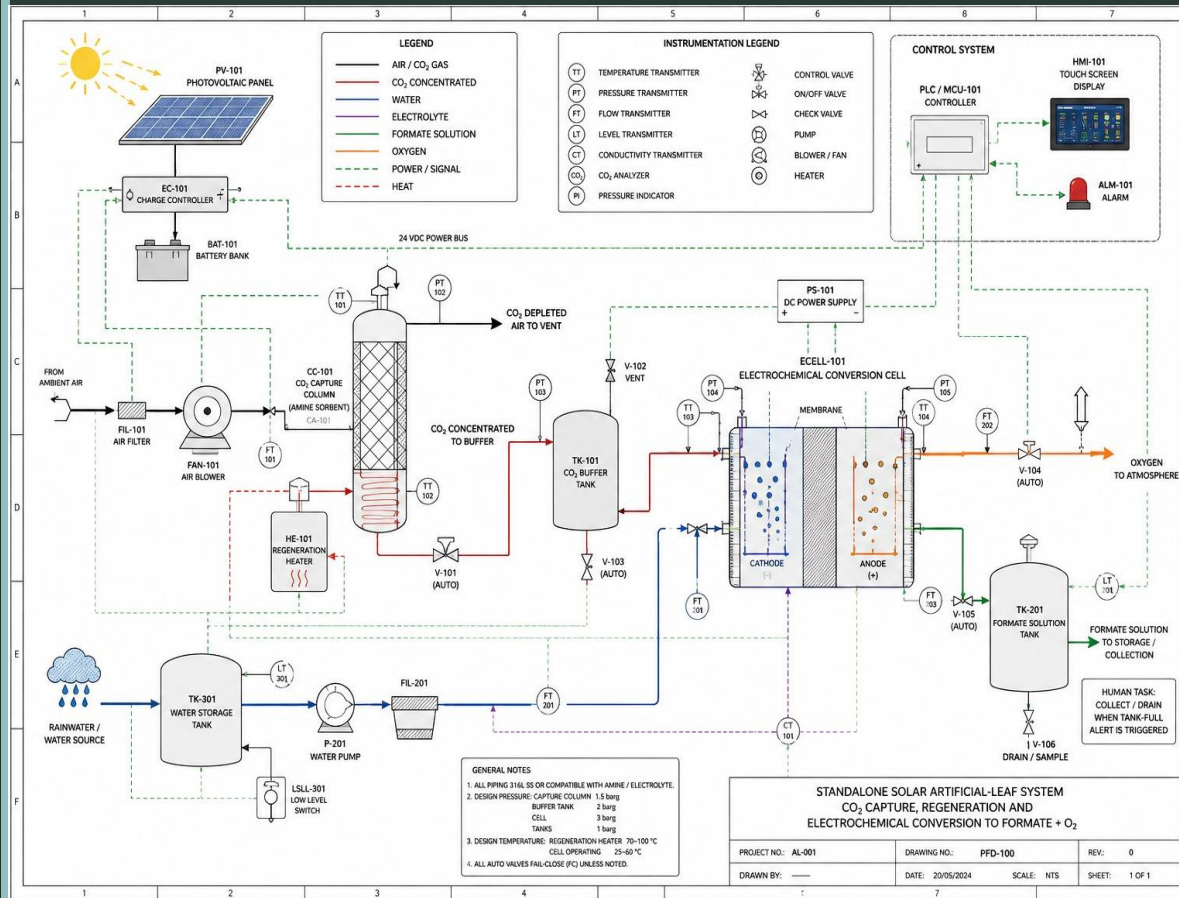
3. Carbon dioxide Conversion

Concentrated carbon dioxide enters the electrochemical cell where it is converted to formate. Water is split to produce oxygen

Outputs

- Oxygen released to the atmosphere
- Formate – a valuable liquid product that can be stored and used.

Design and Operation of SolaForm



Marketability, Cost and Future Impact

Cost of Implementation

Total cost:
R5 876.37

Target Market

-  Schools & Universities
-  Research Institutions
-  Environmental Organisations
-  Future Industrial Applications

Market Outreach Strategy

-  Social Media Marketing
-  School & University Partnerships
-  Science Fairs & Innovation Competitions
-  Research Conferences & Exhibitions
-  Environmental Awareness Campaigns
-  Strategic Partnerships

Future Impact

Capture CO₂
↓
Convert to Formate
↓
Produce Oxygen
↓
Sustainable Future



Project Contributors and Roles



RATSHILAYA KETSEPHILE ROSSA

Role: Research and Documentation

Responsibilities:

Conducted research on artificial photosynthesis, direct air capture, carbon conversion, and formate production. Collaborated with team members in validating and developing the project concept.

Responsible for Folder C documentation

Conducted background research and problem validation.

Developed the marketability and commercialization strategy.

Performed cost analysis and supplier sourcing.



MASEBE REFILOE PRIMROSE

Role: Systems Development & Software Lead

Responsibilities:

Refined and transformed the SolaForm concept into a practical and implementable solution.

Defined the operating stages and system workflow.

Responsible for Folder A: Source Code.

Developed the Wokwi simulation and software implementation.

Documented programming methods, frameworks, and control strategies.

Validated system functionality through simulation and testing.



MALEKA MASHEGO NATHAN

Role: Concept Developer & Design Lead

Responsibilities:

Conceived and initiated the SolaForm Artificial Leaf concept.

Identified the environmental challenge of rising atmospheric CO₂ levels.

Proposed the artificial photosynthesis approach for carbon capture and formate production.

Responsible for Folder B documentation

Conducted and documented simulation designs and testing.

Conclusion

SolaForm Artificial Leaf demonstrates how atmospheric CO₂ can be captured and converted into formate while producing oxygen using solar energy.

- Carbon Capture
- Formate Production
- Oxygen Generation
- Sustainable Innovation



“SolaForm transforms carbon dioxide into valuable products, supporting a cleaner and more sustainable future.” 🌱 ☀️ ♻️